Decisions Log

This document records key decisions made during the EMS Readiness Optimization project.

Decision Template

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### [DEC-XXX] Decision Title

**Date**: YYYY-MM-DD
**Decision Maker**: [Name/Role]
**Status**: Proposed / Accepted / Superseded

**Context**:
[What situation prompted this decision?]

**Options Considered**:
1. Option A: [Description]
2. Option B: [Description]
3. Option C: [Description]

**Decision**:
[Which option was chosen and why]

**Consequences**:
[What are the implications of this decision?]
```

Decisions

[DEC-001] Study Area Selection

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select primary and secondary study areas for the EMS optimization analysis.

Options Considered:
1. All of NYC: Comprehensive but computationally expensive
2. Manhattan only: Focused, manageable scope
3. Manhattan + CBD robustness check: Balanced approach

Decision:
Option 3 - Manhattan as primary study area with CBD (MTA Congestion Relief Zone) as secondary area for robustness testing.

Consequences:
- Allows focused analysis with reasonable computational requirements
- CBD provides natural robustness check for high-demand area
- Results may not generalize to outer boroughs

[DEC-002] Staging Location Candidates

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to define candidate locations for EMS staging.

Options Considered:
1. Grid-based locations: Uniform coverage
2. Hospital locations: Practical operational sites
3. FDNY firehouses: Existing infrastructure
4. Combined approach: Multiple location types

Decision:
Option 3 - Use FDNY firehouses as staging candidates.

Consequences:
- Leverages existing infrastructure
- Realistic operational scenario
- Limited to current firehouse locations (may not be optimal in absolute terms)

[DEC-003] Simulation Framework

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select simulation methodology for EMS system modeling.

Options Considered:
1. Agent-based simulation: Detailed but complex
2. Discrete-event simulation (DES): Standard for queueing systems
3. System dynamics: Too aggregated for this application

Decision:
Option 2 - Discrete-event simulation using SimPy.

Consequences:
- Well-suited for modeling arrival, dispatch, service processes
- Efficient computation
- Python ecosystem enables integration with optimization

[DEC-004] Optimization Approach

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select optimization method for ambulance staging.

Options Considered:
1. Heuristic approaches: Fast but no optimality guarantee
2. Mixed-integer programming: Optimal but potentially slow
3. Linear programming (relaxed): Fast, near-optimal for many problems
4. Metaheuristics: Flexible but tuning required

Decision:
Option 3 - Linear programming using PuLP, with integer constraints if needed.

Consequences:
- Fast solution times
- Optimality guarantees (for LP formulation)
- May need to round/adjust for integer ambulance counts

[DEC-005] Data Temporal Scope

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to determine which years of crash data to use.

Options Considered:
1. Most recent year only: Current patterns
2. Last 3 years: Balances recency and stability
3. All available data: Maximum sample size

Decision:
Option 3 - Use all available data for demand pattern estimation, with options to filter by date range.

Consequences:
- Maximum statistical power for pattern estimation
- May include outdated patterns (e.g., pre/post COVID)
- Can subset for sensitivity analysis

[DEC-008] Alternative Distance Metric — Manhattan (Taxicab) Distance

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Haversine (great-circle) distance underestimates true road distances in a grid-based street network

like Manhattan. Need to evaluate whether a Manhattan (taxicab) distance metric produces materially different optimization and simulation results.

Options Considered:
1. Haversine only (status quo)
2. Manhattan (taxicab) distance replacing Haversine
3. Both metrics compared side-by-side with simulation evaluation
4. Road network distance (OSRM / Google routing)

Decision:
Option 3 — Implement Manhattan distance as an alternative metric, run P2 optimization with both, and compare via simulation.

Rationale:
- Manhattan distance is ~27% longer on average than Haversine for Manhattan geography, better approximating grid-based street travel
- Side-by-side comparison quantifies sensitivity to metric choice
- Road network routing (Option 4) out of scope — requires external APIs and adds complexity without changing relative policy rankings
- Simulation dispatcher always uses Haversine internally, so this isolates the effect on allocation decisions

Consequences:
- Added `manhattan_distance()` function to `distance.py` and `metric` parameter to `build_distance_matrix()`
- Manhattan matrix generated: mean 5.48 mi vs. Haversine 4.29 mi
- Experiment showed near-identical simulation results (mean RT 2.55 min for both), confirming P2 allocation is robust to distance metric choice
- Only 2 of 48 firehouses differ between allocations
- Validates that Haversine is a sufficient approximation for this study

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[DEC-009] CBD-Focused vs. Manhattan-Wide Optimization

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
The CBD (precincts 1, 5, 6, 7, 9, 10, 13, 14, 17, 18) accounts for ~56% of total demand. Need to evaluate whether a CBD-focused optimization strategy (up-weighting CBD demand) improves CBD response times and is preferable to the Manhattan-wide P2 model.

Options Considered:
1. Manhattan-wide P2 only (status quo)
2. CBD-only model (ignore non-CBD precincts)
3. CBD-focused model with 3× demand weight on CBD precincts
4. Multi-objective (Pareto) CBD/non-CBD optimization

Decision:
Option 3 — Implement CBD-focused demand-weighted model (3× CBD weight) and compare against Manhattan-wide P2 via simulation.

Rationale:
- CBD-focused (3× weight) tests whether concentrating resources improves CBD performance
- Comparison with Manhattan-wide P2 quantifies equity trade-offs
- CBD-only model (Option 2) is impractical — abandons non-CBD coverage
- Multi-objective Pareto (Option 4) deferred to future work

Consequences:
- Added `build_cbd_focused_demand_weighted()` and `build_cbd_focused_coverage()` to `models.py`
- CBD-focused P2 does NOT improve CBD RT (2.50 vs. 2.47 min, +1.2%) but dramatically worsens non-CBD RT (6.88 vs. 2.66 min, +159%)
- Non-CBD 8-min coverage drops from 100% to 73.6%
- Validates that Manhattan-wide P2 already near-optimally serves the CBD without sacrificing equity
- Recommends against CBD-focused optimization for operational deployment

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[DEC-010] Firehouse Capacity Constraint — cap=2 Default

Date: 2026-03-15
Decision Maker: Technical Lead
Status: Accepted

Context:
The original formulation used cap=5 (DEC-005). Full capacity sensitivity analysis across cap ∈ {1, 2, 3, 5, unlimited} and K ∈ {10, 15, 20, 25, 30, 35, 40, 45, 48} was conducted to determine a realistic default.

Options Considered:
1. Keep cap=5 (original default)
2. Set cap=2 (tighter constraint)
3. Set cap=1 (strictest — one unit per firehouse)
4. Remove capacity constraint entirely

Decision:
Set default firehouse capacity to 2 (`firehouse_capacity: 2` in `configs/optimization.yaml`).

Rationale:
- At K ≤ 30, capacity never binds for any policy (max allocation ≤ 1 unit/FH), so cap=2 and cap=5 yield identical results
- At K=40, cap=2 forces wider geographic dispersion (29 vs. 24 firehouses for P2) with negligible performance loss (<0.15 min mean RT)
- cap=2 better reflects typical FDNY firehouse physical infrastructure than cap=5
- cap=1 is overly restrictive (requires 40 firehouses at K=40, some with suboptimal locations)
- Full sensitivity analysis (450 runs) confirms robustness across all capacity levels

Consequences:
- All Production V2 results use cap=2 as default
- V1→V2 P0 improvement is entirely due to P0-spatial baseline change (not capacity)
- P1 and P2 results unchanged between V1 and V2 (capacity does not bind at typical K)
- Future work can incorporate per-firehouse capacity data from FDNY

[DEC-011] Spatially-Stratified P0 Baseline (P0-spatial)

Date: 2026-03-15
Decision Maker: Technical Lead
Status: Accepted

Context:
The original P0 (uniform allocation) assigned units to firehouses in database index order, which produced geographically clustered allocations biased toward lower Manhattan. This made P0 an artificially weak baseline.

Options Considered:
1. Keep index-based P0 (status quo)
2. Latitude-based spatial stratification (P0-spatial)
3. Random allocation with fixed seed
4. Grid-based stratification (latitude + longitude)

Decision:
Replace index-based P0 with latitude-based spatially-stratified P0 (P0-spatial) as the standard baseline.

Rationale:
- Index-based P0 is not a meaningful “uniform” allocation — it depends on arbitrary database ordering
- Latitude-based stratification divides Manhattan into equal-width bands and round-robins units, ensuring even north–south coverage
- Manhattan’s elongated geometry makes latitude the dominant spatial axis; adding longitude stratification adds complexity with minimal benefit
- Random allocation is non-reproducible without seed management
- P0-spatial at K=20: mean RT = 3.17 min vs. 8.08 min for index-based P0 (−60.7%)
- P0-spatial provides a fairer baseline for evaluating P1 and P2 improvements

Consequences:
- P2 improvement over P0 narrows from 68% to 19% (2.57 vs. 3.17 min) — reflects that much of P0’s original weakness was geographic clustering, not lack of demand-weighting
- All Production V2 results use P0-spatial
- Added `spatially_stratified_allocation()` and `spatial_stratification_analysis()` to `optimization/policies.py`
- Historical V1 results preserved in `results/production_v1/` for comparison

Last Updated: March 15, 2026

[DEC-003] Optimization Policy Selection

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Multiple optimization formulations exist for EMS allocation. Need to determine which models to implement and compare for Manhattan EMS readiness.

Options Considered:
1. Demand-Weighted Optimization: Minimize demand-weighted average response time
2. P-Median: Minimize total response time (unweighted)

3. Maximal Coverage: Maximize demand covered within threshold
4. Set Covering: Cover all demand with minimum units
5. Hybrid: Multi-objective optimization

Decision:
Implement and compare options 1-3 (Demand-Weighted, P-Median, Maximal Coverage) plus two baseline policies (Uniform, Demand-Proportional).

Rationale:
- Demand-Weighted (P2) aligns with equity goals (weight by demand intensity)
- P-Median (P2b) provides comparison for unweighted optimization
- Maximal Coverage (P2c) represents pure coverage objective
- Baselines (P0, P1) provide non-optimized benchmarks
- Set Covering is subsumed by other models
- Multi-objective deferred to future work

Consequences:
- Need to implement 3 MIP solvers + 2 baseline policies
- Compare 5 policies × 4 K values = 20 scenarios
- Provides comprehensive policy comparison for decision-makers

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[DEC-004] Unit Count Scenarios (K)

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Need to determine which unit counts to evaluate for optimization and sensitivity analysis.

Options Considered:
1. K = {20, 30, 40, 48} - Current recommendation
2. K = {10, 20, 30, 40, 50} - Wider range
3. K = {25, 30, 35, 40, 45, 50} - Finer granularity
4. K = 40 only - Current Manhattan capacity

Decision:
Use K = {20, 30, 40, 48}

Rationale:
- K=20: Resource-constrained scenario (50% of current)
- K=30: Moderate resource scenario (75% of current)
- K=40: Current Manhattan capacity (baseline)
- K=48: All firehouses staffed with 1 unit (upper bound given capacity=5)
- Provides sufficient range to observe diminishing returns
- Computationally efficient (4 scenarios × 5 policies = 20 optimizations)

Consequences:
- May miss optimal K if it falls outside range (e.g., K=25)
- But results show K=20 achieves near-optimal for P2, so range is sufficient
- Future work can refine with continuous K if needed

[DEC-005] Firehouse Capacity Constraint

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Need to determine maximum units that can be allocated to a single firehouse.

Options Considered:
1. No limit (unconstrained)
2. Capacity = 3 units per firehouse
3. Capacity = 5 units per firehouse
4. Capacity = 10 units per firehouse
5. Variable capacity by firehouse size

Decision:
Use capacity = 5 units per firehouse (uniform across all locations)

Rationale:
- Based on FDNY operational guidance (typical firehouse can support 4-6 units)
- Prevents unrealistic concentration (e.g., all 40 units at one location)
- 5 units allows sufficient flexibility for optimized allocation
- Uniform capacity simplifies optimization formulation
- Conservative estimate (actual capacity varies 3-10 by location)

Consequences:
- P2 allocates up to 5 units at high-demand firehouses (Midtown, Financial District)
- May under-utilize capacity at larger firehouses
- Future work can incorporate actual capacity data per location

[DEC-006] Coverage Threshold for Maximal Coverage Model

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Maximal Coverage model requires threshold τ (minutes) to define “covered” demand.

Options Considered:
1. $\tau = 5$ minutes (aggressive response target)
2. $\tau = 8$ minutes (NFPA recommendation)
3. $\tau = 10$ minutes (relaxed target)
4. $\tau = 6$ minutes (NYC local law)
5. Multiple thresholds for sensitivity analysis

Decision:
Use $\tau = 8$ minutes (NFPA standard)

Rationale:
- NFPA 1710 specifies 8 minutes for 90% of emergencies
- Aligns with national standards for comparability
- NYC local law (6 min) is aspirational but not consistently achievable
- All policies achieve 100% coverage at $\tau=8$ with $K\geq20$
- Using $\tau=5$ would require more units; $\tau=10$ is too lenient

Consequences:
- P2c (Maximal Coverage) achieves 100% coverage but with worse response time (3.48 min)
- May want to test $\tau=6$ in future work to align with NYC requirements
- 8-minute threshold appropriate for Manhattan context (dense, short distances)

[DEC-007] Recommended Policy for Deployment

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Proposed

Context:
After comparing 5 policies, need to recommend best allocation strategy for Manhattan EMS.

Options Considered:
1. P0 (Uniform): Equal distribution
2. P1 (Demand-Proportional): Simple heuristic
3. P2 (Demand-Weighted): Minimizes weighted response time
4. P2b (P-Median): Minimizes total response time
5. P2c (Maximal Coverage): Maximizes coverage

Decision:
Primary recommendation: P2 (Demand-Weighted) with K=20-30

Rationale:
- Best response time: 2.49-2.54 min (0-2% from optimal)
- 100% coverage at K≥20
- Efficient: Uses only 20-23 firehouses vs. 34+ for other policies
- Significant improvement over P0: 17-86% faster response time
- Fast solve time: <0.05 seconds
- Diminishing returns beyond K=30 (additional units provide <0.01 min benefit)

Alternative: P2b for political feasibility (distributes units more evenly)

Consequences:
- Requires MIP solver integration (PuLP + CBC)
- Concentrated allocation may face political resistance (some firehouses get 0 units)
- Recommend phased implementation: Start with P1, transition to P2
- Quarterly reallocation based on updated demand patterns